Contractivity Cannot Be Removed from the Approximation-Number Truncation Theorem

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Status. Candidate counterexample, likely valid, for human review.

1 Source question

Deepesh's Theorem 4 states the following. Let Y be a dual space, $T \in B(X, Y)$, and $P_n \in B(X)$, $Q_n \in B(Y)$ satisfy $\|P_n\| \, \|Q_n\| \leq 1$. If $T_n = Q_n T P_n$ converges to T in the weak-star operator topology, then $a_k(T_n) \to a_k(T)$ for every $k \in \mathbb{N}$ [1, Theorem 4].

Section 4 asks whether this conclusion still holds when the norm restrictions on P_n and Q_n are removed [1, Section 4, PDF pp. 13–14]. The exact source passage spans two pages:

> **4 Concluding remarks and potential research problems**
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> Although the convergence problem has been addressed for approximation numbers, it is not known if the result in Theorem 4 holds when the norm restrictions on P_n and Q_n are removed. This is a significant case to consider, as in Banach spaces, it is not always
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> possible to obtain operators of norm 1 or less that converge to the identity operator. Also, for the other s-numbers, only partial results were achieved by applying Theorem 4, as the truncations considered in the results here are not two sided. It would be of considerable interest to obtain similar convergence results for other s-numbers, either by modifying Theorem 4 or through alternative approaches.

The first sentence of the passage is the target here; the subsequent request about other s-numbers is a separate research direction.

2 Definition

For $S \in B(X, Y)$, the kth approximation number is

$$a_k(S) = \inf\{\|S - F\| : F \in B(X, Y), \ \operatorname{rank} F < k\}.$$

3 Proof intuition

Strong convergence only tests each fixed vector, so it cannot see a block whose location drifts to infinity. We use finite-rank truncations that are correct on the first n coordinates, double the next n coordinates, and discard the rest. The doubled block moves outward, so the truncations converge strongly to the identity. But its dimension grows: for any fixed k, every rank-less-than-k approximant misses a nonzero direction in that block. Along that direction the truncation still has size 2, forcing the kth approximation number to remain 2 rather than converge to the value 1 for the identity.

4 Counterexample

Let $H = \ell_2(\mathbb{N})$ with standard orthonormal basis $(e_j)_{j\geq 1}$, and let R_m denote the orthogonal projection onto $\operatorname{span}\{e_1, \ldots, e_m\}$. For $n \geq 1$, set

$$P_n = R_{2n}, \qquad E_n = R_{2n} - R_n, \qquad Q_n = R_n + 2E_n,$$

and take $X = Y = H$ and $T = I_H$.

Theorem 1. *The operators P_n and Q_n have finite rank and both converge strongly to I_H. Moreover, if $T_n = Q_n T P_n$, then $T_n \to T$ strongly (hence in the weak-star operator topology), but for every fixed $k \in \mathbb{N}$,*

$$a_k(T_n) = 2 \quad (n \geq k), \qquad \textit{whereas} \qquad a_k(T) = 1.$$

Consequently the norm restriction in Theorem 4 cannot be removed, even for finite-rank truncation factors that themselves converge strongly to the identity.

Proof. Both P_n and Q_n have range contained in $R_{2n}H$, hence have finite rank. Clearly $P_n x \to x$ for every $x = (x_j) \in H$. The three mutually orthogonal pieces R_nH, E_nH, and $(I - R_{2n})H$ show that

$$(Q_n - I)x = E_n x - (I - R_{2n})x$$

and therefore

$$\|(Q_n - I)x\|^2 = \sum_{j=n+1}^{2n} |x_j|^2 + \sum_{j>2n} |x_j|^2 = \sum_{j>n} |x_j|^2 \longrightarrow 0.$$

Thus $Q_n \to I$ strongly. Since $Q_n R_{2n} = Q_n$,

$$T_n = Q_n I_H P_n = Q_n,$$

so $T_n \to I_H = T$ strongly. Because H is canonically a dual Hilbert space, strong convergence implies the weak-star operator convergence required by the source theorem. Notice also that

$$\|P_n\| = 1, \qquad \|Q_n\| = 2,$$

so the removed product restriction is violated by the fixed factor 2.

We next compute the approximation numbers directly. The zero operator gives $a_k(I_H) \leq 1$. Conversely, if F has rank less than k, then $\ker F$ is nonzero (indeed infinite dimensional). For a unit vector $x \in \ker F$,

$$\|(I_H - F)x\| = 1,$$

so $a_k(I_H) = 1$.

Now fix k and take $n \geq k$. Since Q_n acts as $2I$ on the n-dimensional space E_nH, its norm is 2, and hence $a_k(Q_n) \leq 2$. For any F of rank less than k, the restriction $F|_{E_nH}$ has rank less than $k \leq n = \dim(E_nH)$. It therefore has a nonzero kernel vector. Choosing a unit vector $x \in E_nH$ with $Fx = 0$ gives

$$\|(Q_n - F)x\| = \|2x\| = 2.$$

Thus $\|Q_n - F\| \geq 2$ for every admissible F, so $a_k(Q_n) = 2$. Since $T_n = Q_n$, this proves the displayed failure for every fixed k. $\square$

5 Verification and exact scope

Every assertion is checked algebraically in the proof. In particular, the example is not exploiting an unbounded sequence: $\sup_n \|P_n\| = 1$ and $\sup_n \|Q_n\| = 2$. It also works simultaneously for all fixed k, and both truncation factors are finite rank and strongly convergent to the identity.

The counterexample answers the literal first question in Section 4. It does not address the separate request for two-sided convergence theorems for other s-numbers. It also does not determine which intermediate hypotheses—for example, asymptotic contractivity—might recover the conclusion.

6 Novelty check

The bounded search on 9 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; exact searches for arXiv:2401.08483, its title, and the sentence about removing the norm restrictions; and close searches for approximation-number truncation counterexamples without contractivity. We checked both arXiv v2 and the official 2025 journal version, which still prints the question as open. No later exact answer or matching counterexample was found. This supports, but cannot certify, novelty.

7 Human-review recommendation

Accept as a likely valid negative answer after checking that "the norm restrictions on P_n and Q_n are removed" is intended literally relative to Theorem 4. The finite-rank, strong-convergence construction also satisfies the natural truncation conditions suggested by the source's motivation.

References

[1] K. P. Deepesh, *Approximation results on s-numbers of operators*, arXiv:2401.08483v2 (2024); published in *Palestine Journal of Mathematics* **14**(3) (2025), 308–318, `https://arxiv.org/abs/2401.08483`.